

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement provides the complete R21 continuum, price-sensitivity, policy-improvement, and cost ledgers and identifies all 136 frozen-policy replays. It preserves the complete R19, R18, and R14 supplemental development, including historical feedback failures and all high-dimensional comparisons. Superseded introductory and concluding exposition is reproduced verbatim in dated historical annexes.

1. R21 EVIDENCE IDENTITIES AND REPRODUCTION

The initial referee input is the R18 report at review commit 074849b9aad1812b59e25e1d3833383ed11aa401. The manuscript inherits the R20 source head 6916f250ca2115399dccf2df125c0b13f304547b. The R20 confirmatory protocol was committed before execution at 238ecc0052dc418e1fe6fd06c3daa4a93da1a335; its successful scientific run is 35802793602 at source b1a2ea39d895b330e5a11b4e55d86b7a4985b101. These identifiers distinguish protocol, training, prior paper, new analysis, and publication. The publication manifest does not attempt to contain its own commit hash.

R21 adds continuum proofs and independently executes frozen-policy checks. It preserves each R20 actor, neural-weight file, arithmetic enclosure, acceptance decision and resource record without alteration. Source and result SHA256 identities are recorded separately. No new training randomness is claimed. The final payoff interval is the hull of the inherited and replayed enclosures, including both arithmetic implementations; taking a more favorable intersection is unnecessary.

1.1. *Complete accuracy, price, and cost ledgers*

The machine-readable file results/continuum_audit.json contains every seed and retained level, the Jensen upper bound, uniform payoff gain, true-regret contraction, directed Hessian bounds, analytic admissibility constants, and every target hit or miss. The file results/replay_summary.json identifies every new execution and interval-overlap test. Overlap is an arithmetic consistency test; validity also depends on the shared analytic proof in the main appendix.

TABLE I

FIRST RETAINED .005 HIT ON K . GENERATION AND CHECKING ARE MEASURED SECONDS; SHARED-UPPER AND ADDITIONAL PUBLICATION AUDITS ARE CHARGED SEPARATELY IN THE TEXT.

Method	Budget	Generation	Checking	Sum	RSS (KiB)
Seed 20100	100	1.107	15.036	16.143	427908
Seed 20101	100	1.096	15.024	16.119	427916
Seed 20102	100	1.099	15.083	16.181	427940
Seed 20103	100	1.084	15.023	16.107	428008
Seed 20104	100	1.074	15.060	16.133	429192
Direct	10	0.153	15.360	15.512	n.r.

TABLE II

FIXED-POLICY PRICE TRANSPORT: ENDPOINT BOUNDS UNIFORM OVER ALL FIVE NEURAL MIXTURES AND ALL INITIAL STATES IN K . ENDPOINTS BOUND EACH ENTIRE INTERVENING SEGMENT.

Adjustment price k	Regret upper bound
1.5	0.00332694
1.7490234375	0.00298659
2.0	0.00290848
2.0830078125	0.00288265
2.423828125	0.00313176
2.5	0.00326455

At tolerance .002, every seed and the direct stochastic comparator miss the reported target. At .01 and .005, the first retained hitting levels coincide: 100 updates per neural expert and 10 iterations per direct expert. A final-study wall clock includes work after the first hit; it is not substituted for time to a declared tolerance. Conversely, a sum of measured generation and checking times is identified as that sum, rather than mislabeled process wall time. Full R20 resource files retain the actual wall clocks. The shared twenty-node dual-library construction costs 415.789 seconds after upward rounding. Cold-start comparisons add it to every method or state explicitly that it is reused. Final MPFR and R21 publication checks are separately charged audits. Peak memory absent from the original direct run is explicitly missing, not estimated from parameter counts.

1.2. *What is reproduced and what is preserved*

The new replay consists of 80 neural expert checks, 20 final-expert MPFR checks, 16 direct-stochastic checks, and 20 compensated-policy checks: 136 executions. The first group includes initialization and all three retained proposals for each of twenty experts. The MPFR processes are isolated from the binary processes to avoid sharing the loaded interval module. Every reported actor hash is checked. The script fails on a non-overlapping price, reserve, portfolio, or payoff interval. Generated tables are derived from directed calculations and round in the conservative direction.

The inherited record retains the all-ten-seed actor-critic factorial, 30 trace checkpoints, signed classical decompositions, continuation-lift interventions, and high-dimensional learned-gradient/correction study. These are not counted among the new 136 replays. The original all-domain neural error of approximately 7.3, the width-dependent architecture reversal, and accelerated-gradient advantages at stringent nonlinear-control tolerances remain visible. The corrected localized success does not remove any of those observations.

1.3. *Execution commands*

From the repository root, run the new checker and continuum calculation, then build the three documents:

```
R21=revisions/2026-09-23-r21/replication
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
python "$R21/replay.py"
python "$R21/audit.py"
python "$R21/representation_audit.py"
python "$R21/build.py"
```

The original R20 training commands and frozen protocol remain in their original directory. Re-running them is a fresh execution, not a prerequisite for reading the already serialized policies. Their outputs should be directed to a separate copy when preserving the publication snapshot.

2. READING THE RETAINED DEVELOPMENT

The remaining supplement and the retained sections of the main paper reproduce earlier substantive mathematical and numerical material. References to “new” runs within those retained sections mean new in their own R14–R19 revision, not additional R21 training. The historical introductions and conclusions are reproduced verbatim below so that revised exposition does not silently delete prior material. Current headline results and scope are those of the R21 introduction, stochastic-policy section, and conclusion.

3. FURTHER INTERMEDIATE-SNAPSHOT REPORT AND REPRESENTATION AUDIT

The additional report at commit `bc118f20f6361cdae668141e45f64c436ee6a2a` reviewed the earlier source-archive snapshot, not the present R21 proofs. The combined response addresses both reports. The new main-text restricted-Markov proposition identifies the exact feedback implemented by the decoder. The representation audit verifies all twenty adverse vertex payoff comparisons and equality of initial parameter dictionaries across vertices within each of five seeds. The old root index is preserved in the history directory; every other inherited file remains at its original path.

4. R19 EXECUTION, SCOPE, AND COMPLETE DIAGNOSTICS

The R19 protocol was committed before the new actor training and the new finite-query study. The actor checkpoints are 0, 50, 200, 800 for seeds 19100–19104. All checkpoint networks, their exact compiled dyadic policies, payoff intervals, acceptance records, and resource logs are retained. The initial and final policies enter the coupled-payoff proof; its deterministic state-extension and wealth-compensation calculations are subsequent analytical consequences, not additional randomized experiments. The attribution study reuses historical frozen networks and is expressly not relabeled as new training.

The primary reproduction evaluates the independent payoff gate before the next Adam update. The optimizer state and network are restored on rejection. An acceptance requires a strictly positive lower endpoint for new payoff minus incumbent payoff; a smaller optimality bound alone is insufficient. All 15 noninitial proposals pass. A separate audit verifies the observable progress test at $\kappa = .02$ for every retained transition, $\kappa = .75$ for each reference-state initial-to-final pair, and the final-to-initial true-regret ratio below .42 on K . These are executed-trajectory statements, not an assumption of global optimizer convergence.

4.1. *All upper-face trajectories*

Tables III and IV report every accessible historical checkpoint, including the signed positive comparison term and the fixed-witness lower floor. Their differences are not treated as actor-value differences. All ten actor–critic interchanges are in the main text. The positive/negative decomposition remains part of the numerical claim, rather than an optional diagnostic hidden behind a total bound.

TABLE III
CONTINUATION TRACE TRAJECTORIES, PART 1

Seed	Update	Trace lower	Trace upper	Floor lower	B^+
17100	0	6.679868	6.679869	0.000000	16.983516
17100	100	1.755992	1.755993	4.616272	9.746178
17100	400	0.094970	0.094971	6.277294	7.339625
17101	0	6.661007	6.661008	0.000000	17.020635
17101	100	0.245042	0.245043	6.127222	7.547735
17101	400	0.053474	0.053475	6.318790	7.281386
17102	0	6.686390	6.686391	0.000000	16.975928
17102	100	1.820291	1.820292	4.551973	9.840285
17102	400	0.097382	0.097383	6.274882	7.343312
17103	0	6.648314	6.648315	0.000000	17.006121
17103	100	0.235585	0.235586	6.136679	7.535185
17103	400	0.052760	0.052761	6.319505	7.280303
17104	0	6.694775	6.694776	0.000000	17.003002
17104	100	1.964101	1.964102	4.408163	10.044977
17104	400	0.093338	0.093339	6.278926	7.336434

All 30 accessible checkpoints are included. A fixed-witness floor is not a lower bound on actual policy regret.

TABLE IV
CONTINUATION TRACE TRAJECTORIES, PART 2

Seed	Update	Trace lower	Trace upper	Floor lower	B^+
17105	0	6.675560	6.675561	0.000000	17.035163
17105	100	0.252011	0.252012	6.120253	7.555938
17105	400	0.052751	0.052752	6.319513	7.280018
17106	0	6.666272	6.666273	0.000000	16.981173
17106	100	1.565681	1.565682	4.806583	9.477079
17106	400	0.097374	0.097375	6.274890	7.343666
17107	0	6.682934	6.682935	0.000000	17.099704
17107	100	0.286118	0.286119	6.086146	7.604406
17107	400	0.051775	0.051776	6.320489	7.278320
17108	0	6.653260	6.653261	0.000000	17.019389
17108	100	1.587040	1.587041	4.785224	9.503279
17108	400	0.093895	0.093896	6.278369	7.338342
17109	0	6.694382	6.694383	0.000000	17.076539
17109	100	0.250573	0.250574	6.121691	7.553445
17109	400	0.052498	0.052499	6.319766	7.279140

All 30 accessible checkpoints are included. A fixed-witness floor is not a lower bound on actual policy regret.

4.2. Classical feedback decomposition

The finest-grid deterioration is associated with a positive policy-side component, absent at the intermediate grids. The arithmetic decomposition identifies where the reported upper bound changes. It does not establish that true policy value worsens, and it does not separate interpolation error from witness mismatch without an additional fixed-policy solve. The new SLSQP frontier is a time-control comparison, not a substitute label for these feedback methods.

TABLE V
SIGNED DECOMPOSITION OF UNRESTRICTED CLASSICAL FEEDBACK CANDIDATES

Candidate	B^+	B^-	Total
MC-17	7.314510	0.000000	7.314510
MC-25	7.282120	0.481344	7.763463
MC-9	7.339816	0.000000	7.339816
SL-17	7.314559	0.000000	7.314559
SL-25	7.286608	0.487515	7.774123
SL-9	7.342251	0.000000	7.342251

The medium-to-fine deterioration coincides with a positive policy-side term. This separates the terms in the reported bound, but by itself does not identify interpolation error, witness fit, or true policy deterioration.

4.3. All learned warm-start frontiers

The following tables retain all three neural seeds and four tolerances for each dimension. The paired classical method uses the same query vectors. Solver time includes neural inference where applicable; check time is the final MPFR certificate. No finite accounting break-even is returned when the measured saving is nonpositive. Offline training costs are the recorded historical costs, so break-even counts are transferred-cost estimates rather than a matched-hardware retraining study. Negative or absent amortization outcomes are retained. The learned-gradient budgets

TABLE VI
ONLINE WORK AND CONDITIONAL AMORTIZATION AT DIMENSION 8

Seed	Target	NN ms	AG ms	Q_{solve}	Q_{all}	GD bound N/Z
17400	10^{-2}	1.696	3.918	2531	1960	4/9
17401	10^{-2}	1.651	3.918	2596	2361	4/9
17402	10^{-2}	1.725	3.918	3043	4194	4/9
17400	10^{-3}	2.757	4.940	2576	2255	7/12
17401	10^{-3}	2.660	4.940	2580	2586	7/12
17402	10^{-3}	2.729	4.940	3018	5234	7/12
17400	10^{-4}	4.274	6.420	2620	2535	10/15
17401	10^{-4}	4.145	6.420	2586	2814	10/15
17402	10^{-4}	4.297	6.420	3144	5174	10/15
17400	10^{-6}	7.652	9.510	3027	2665	16/21
17401	10^{-6}	7.654	9.510	3170	3414	16/21
17402	10^{-6}	7.631	9.510	3553	8837	17/21

Times are mean five-repeat per-query medians. Break-even counts combine the recorded historical training time with this query execution; they are transferred-cost estimates, not matched-hardware or certified speedups. A dash means no positive measured saving. The all-cost count includes final directed checking. The last column is a proved exact-real GD correction budget using the maximum independently enclosed initial gradient over this finite query set; N and Z denote neural and zero initialization.

use a directed maximum over each finite query set in the main paper’s learned-quality theorem. They do not certify the entire cube. The stored floating-point plans are separately checked at the exact stored queries, so an exact-real contraction theorem is not silently used as a floating-point error bound. The complete files contain 1,920 target-check records, including 1,152 neural and 768 classical records; every declared target is attained within the 64-correction cap.

TABLE VII
ONLINE WORK AND CONDITIONAL AMORTIZATION AT DIMENSION 32

Seed	Target	NN ms	AG ms	Q_{solve}	Q_{all}	GD bound N/Z
17400	10^{-2}	2.479	4.977	3037	2537	5/10
17401	10^{-2}	2.343	4.977	3105	2206	5/10
17402	10^{-2}	2.504	4.977	2689	1664	5/10
17400	10^{-3}	3.639	5.903	3351	3269	8/13
17401	10^{-3}	3.613	5.903	3569	2610	8/13
17402	10^{-3}	3.580	5.903	2863	3978	8/13
17400	10^{-4}	5.313	7.489	3485	3010	11/17
17401	10^{-4}	5.282	7.489	3704	2332	11/17
17402	10^{-4}	5.313	7.489	3056	1847	11/17
17400	10^{-6}	9.034	10.796	4305	6124	18/23
17401	10^{-6}	8.903	10.796	4319	2704	17/23
17402	10^{-6}	8.996	10.796	3694	2277	18/23

Times are mean five-repeat per-query medians. Break-even counts combine the recorded historical training time with this query execution; they are transferred-cost estimates, not matched-hardware or certified speedups. A dash means no positive measured saving. The all-cost count includes final directed checking. The last column is a proved exact-real GD correction budget using the maximum independently enclosed initial gradient over this finite query set; N and Z denote neural and zero initialization.

TABLE VIII
ONLINE WORK AND CONDITIONAL AMORTIZATION AT DIMENSION 128

Seed	Target	NN ms	AG ms	Q_{solve}	Q_{all}	GD bound N/Z
17400	10^{-2}	5.748	6.577	13535	7216	11/12
17401	10^{-2}	5.748	6.577	16056	8768	11/12
17402	10^{-2}	5.743	6.577	16313	8124	11/12
17400	10^{-3}	7.545	7.518	—	—	14/15
17401	10^{-3}	7.533	7.518	—	239322	14/15
17402	10^{-3}	7.518	7.518	41296137	138022	14/15
17400	10^{-4}	9.411	9.383	—	46790	17/18
17401	10^{-4}	9.436	9.383	—	62773	17/18
17402	10^{-4}	9.464	9.383	—	42578	17/18
17400	10^{-6}	13.985	13.122	—	—	23/25
17401	10^{-6}	13.997	13.122	—	—	23/25
17402	10^{-6}	13.992	13.122	—	—	23/25

Times are mean five-repeat per-query medians. Break-even counts combine the recorded historical training time with this query execution; they are transferred-cost estimates, not matched-hardware or certified speedups. A dash means no positive measured saving. The all-cost count includes final directed checking. The last column is a proved exact-real GD correction budget using the maximum independently enclosed initial gradient over this finite query set; N and Z denote neural and zero initialization.

4.4. Reproduction and cost accounting

The entry point is `revisions/2026-09-23-r19/replication/run_all.py`. It runs policy generation and inline acceptance, all-seed attribution, the matched-tolerance query study, deterministic state and welfare extensions, table generation, and validation. The paper build is separate from numerical execution, so a LaTeX success is not reported as scientific gate

closure. The publication workflow records immutable input, protocol, source, and publication objects separately.

Timing records identify generation, policy-value evaluation, query correction, final gradient checking, shared optimal-upper generation/verification, and peak process memory. Shared upper-comparator costs are retained from their historical execution and are not called zero-cost or newly timed. End-to-end neural first-attainment timing includes the earlier retained policy-value checks. Peak memory is process-wide and includes the runtime and loaded libraries; it is not falsely presented as network parameter memory.

All prior tracked files are checked for preservation against the R18 review commit. New arithmetic checks validate the MPFR primitive layer, directed policy ordering, work-count consistency, and stored-plan hashes. They do not constitute an independent formal proof of the model, quadrature, stopping, or comparison theorems. The deployment contract concerns the serialized dyadic controls and checked plans; it does not cover an undocumented online activation implementation or simulator.

5. RETAINED R18 OBJECTIVE AND NUMERICAL DEVELOPMENT

6. CURRENT REVIEW OBJECT AND REPRODUCTION

The current manuscript is `ECTA_R18.tex` with its compiled PDF. This supplement is `SUPP_R18.tex`. The point-by-point response is supplied in both `Mark-down` and `RESPONSE_R18.pdf`. The latest referee input is the 914-line report under `reviews/2026-09-23-econometrica-r16/`. Its immutable review commit is `82af00bc296d`. The reviewed R16 publication commit is `2932b74dad6d`; the R17 scientific source and result commits are `ab79b9db4e7c` and `982326994c85`, respectively.

R18 materializes a new manuscript from those objects and adds the quantitative objective theorem, the finite-horizon defect allocation, the fixed-witness result, the nonlinear-control proof, and the budget-compensation proof. It does not relabel R17 training as newly executed R18 training. The R18 execution checks all 589 immutable scientific files, re-evaluates all 60 original-economy proposal objectives, replays two complete MPFR cylinder certificates, and independently rechecks all 18 final nonlinear neural query plans. It also checks whole-domain control ranges for six unrestricted classical policies and every one of the twenty fresh-library actors, and runs 450 exact-rational arithmetic tests, 80 exact-rational finite-MDP tests, and derivative/Hessian regression diagnostics. Sampling diagnostics are never substituted for the analytical global results.

From the repository root, the new mathematical and numerical checks are reproduced by

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
python revisions/2026-09-23-r18/replication/reproduce.py
pdflatex -halt-on-error ECTA_R18.tex
pdflatex -halt-on-error ECTA_R18.tex
pdflatex -halt-on-error SUPP_R18.tex
pdflatex -halt-on-error SUPP_R18.tex
pdflatex -halt-on-error RESPONSE_R18.tex
pdflatex -halt-on-error RESPONSE_R18.tex
```

The R17 protocols and execution scripts retain the commands for regenerating the underlying neural experiments, unrestricted baselines, nonlinear plans, wealth compensation, and scalar Riccati policies from their declared inputs. The R18 entry point is a theorem-and-evidence audit, not a replacement for those longer scientific reproduction commands. Both scopes are separately identified in the publication manifest. The publication workflow builds all PDFs, preserves the build logs, verifies the new artifacts, and checks that no historical scientific file has changed.

6.1. *Numerical environments and interpretation*

The canonical R17 science ran with Python 3.11.16, NumPy 2.3.5, SciPy 1.17.0, and Torch 2.10.0 on the recorded Linux runner. The new local R18 check used Python 3.13.5, the same NumPy/Torch versions, and MPFR 4.2.2. The publication workflow records its own environment independently. Timings are neither pooled across environments nor treated as a neural speed advantage when error levels differ. Stored network weights are exact dyadic inputs; model constants are exact rationals where specified. The directed arithmetic layer does not formally verify the C interface, control derivation, or shared stopping and quadrature arguments.

All numerical table entries are rounded for presentation unless explicitly identified as outward endpoints. Machine-readable JSON retains the exact stored directed endpoint. A printed rounded number should not replace that endpoint in a new certification calculation. The unchanged original-economy target is .01; no R18 audit treats software test success as attainment of that accuracy target.

7. COMPLETE ORIGINAL-ECONOMY NEURAL DECOMPOSITIONS

Tables below retain the signed components and work for every prescribed original-economy checkpoint. Positive and negative entries are the discounted slab maxima entering the certificate. The all-face ablation multiplies the accessibility-aware critic by the additional factor $1 - e^{4(x-2)}$ in its positive lifting term; actor architecture, initialization logic, optimization settings, action comparison, and verification cover are matched. Seeds of even parity use width 16; odd parity uses width 32. The exact cell record is retained for every entry.

TABLE IX
COMPLETE ACCESSIBILITY-AWARE RECORD

Seed	Updates	Positive	Negative	Total	Train sec.	Audit sec.
17100	0	16.983515	9.316999	26.300514	0.00	12.15
17100	100	9.746177	0.358090	10.104267	17.64	23.76
17100	400	7.339624	0.000000	7.339624	68.49	35.05
17101	0	17.020635	9.479196	26.499830	0.00	31.72
17101	100	7.547734	0.000000	7.547734	26.90	64.64
17101	400	7.281385	0.000000	7.281385	102.01	90.84
17102	0	16.975928	9.350446	26.326374	0.00	10.43
17102	100	9.840284	0.397151	10.237435	18.51	21.81
17102	400	7.343312	0.000000	7.343312	64.26	31.43
17103	0	17.006120	9.505096	26.511217	0.00	27.48
17103	100	7.535185	0.000000	7.535185	21.55	47.47
17103	400	7.280303	0.000000	7.280303	77.80	65.27
17104	0	17.003002	9.364486	26.367488	0.00	7.70
17104	100	10.044976	0.546666	10.591643	12.83	14.51
17104	400	7.336433	0.000000	7.336433	49.19	21.77
17105	0	17.035163	9.433132	26.468295	0.00	21.48
17105	100	7.555938	0.000000	7.555938	14.43	43.38
17105	400	7.280018	0.000000	7.280018	54.39	59.74
17106	0	16.981173	9.371735	26.352908	0.00	5.52
17106	100	9.477079	0.224054	9.701133	8.35	11.19
17106	400	7.343665	0.000000	7.343665	34.17	16.81
17107	0	17.099704	9.567024	26.666728	0.00	16.74
17107	100	7.604406	0.000000	7.604406	12.32	33.22
17107	400	7.278319	0.000000	7.278319	47.39	49.57
17108	0	17.019388	9.320582	26.339971	0.00	5.66
17108	100	9.503279	0.242000	9.745279	7.21	11.41
17108	400	7.338341	0.000000	7.338341	31.20	17.13
17109	0	17.076539	9.450934	26.527472	0.00	17.16
17109	100	7.553444	0.000000	7.553444	13.17	33.16
17109	400	7.279140	0.000000	7.279140	36.65	42.10

Audit seconds are cumulative through the stated checkpoint. All entries have zero skipped or failed cells and zero signed trace budget. These are raw proposals, all retained.

TABLE X
COMPLETE ALL-FACE RECORD

Seed	Updates	Positive	Negative	Total	Train sec.	Audit sec.
17100	0	21.032537	9.283830	30.316368	0.00	12.18
17100	100	10.730580	0.341597	11.072177	18.50	23.70
17100	400	7.345561	0.000000	7.345561	71.28	34.53
17101	0	21.069026	9.445637	30.514662	0.00	34.68
17101	100	7.637592	0.000000	7.637592	24.99	67.45
17101	400	7.263538	0.000000	7.263538	103.12	92.60
17102	0	21.023728	9.317186	30.340914	0.00	11.29
17102	100	10.732082	0.334124	11.066206	18.02	23.40
17102	400	7.351242	0.000000	7.351242	60.06	30.16
17103	0	21.046882	9.471462	30.518344	0.00	27.69
17103	100	7.610538	0.000000	7.610538	20.08	49.96
17103	400	7.263378	0.000000	7.263378	70.77	73.88
17104	0	21.059772	9.331193	30.390965	0.00	7.60
17104	100	10.592155	0.303616	10.895771	12.29	15.33
17104	400	7.345015	0.000000	7.345015	41.50	20.87
17105	0	21.091783	9.399684	30.491466	0.00	22.13
17105	100	7.676889	0.000000	7.676889	15.77	39.53
17105	400	7.261518	0.000000	7.261518	56.39	55.78
17106	0	21.022430	9.338410	30.360841	0.00	5.86
17106	100	10.897741	0.398215	11.295957	8.40	11.60
17106	400	7.349999	0.000000	7.349999	34.32	17.05
17107	0	21.172118	9.533247	30.705365	0.00	16.84
17107	100	7.678567	0.000000	7.678567	12.97	33.23
17107	400	7.262408	0.000000	7.262408	49.32	49.61
17108	0	21.053287	9.287400	30.340687	0.00	5.68
17108	100	10.054138	0.108039	10.162177	7.91	11.55
17108	400	7.345879	0.000000	7.345879	31.51	17.36
17109	0	21.152839	9.417439	30.570278	0.00	17.41
17109	100	7.640730	0.000000	7.640730	11.99	30.66
17109	400	7.261511	0.000000	7.261511	35.44	39.74

Audit seconds are cumulative through the stated checkpoint. All entries have zero skipped or failed cells and zero signed trace budget. These are raw proposals, all retained.

8. THE OBJECTIVE AUDIT AND NUMERICAL DISCREPANCY

The new checker rejects any missing, duplicated, reordered, or unaccounted-for cell and any mismatch between a network and the hash in its certificate. It uses the immutable MPFR cell endpoints as inputs to the objective-bridge check, rather than claiming to regenerate all 60 interval certificates. The two designated full-cylinder replays are identified separately. For every object, its proposal and certified logits are retained in a compressed numerical array with its own hash.

The smooth-maximum computation subtracts the largest stored logit before exponentiation and includes one exact zero logit. Its temperature is the exact rational $3/20$. Directed discounted masses and pairwise reductions enclose every summed quantity. The correction in Theorem 8 is computed against the clipped proposal logits because both maxima already include zero. This avoids assigning a spurious discrepancy to a negative residual whose positive part is exactly zero. The ordinary proposal graph is still not itself a proof of enclosure.

Across all objects, the integrated correction ranges from approximately 1.11149×10^{-14} to 1.19078×10^{-13} . The worst-case soft-maximum allowance is at most 1.631868. All 60 directed reconstructions satisfy the envelope implication. This is a check of the sufficient inequality, not proof of a global minimum of the nonconvex training objective. The current witness floors are orders of magnitude larger than the numerical correction.

9. FINITE-BELLMAN CONDITIONING

The stationary auxiliary recurrence has evaluation amplification $2q/(1-q)^2$ and greedy amplification $(1-q)^{-1}$. At the original discount and representative time steps, the values are:

Time steps	Evaluation multiplier	Greedy multiplier	Evaluation defect for .01 term
64	5.12000×10^6	1600.5001	1.95313×10^{-9}
128	2.04800×10^7	3200.5001	4.88282×10^{-10}
256	8.19200×10^7	6400.5001	1.22071×10^{-10}

The last column controls only that additive evaluation term in one use of the recurrence, not its infinite-iteration accumulated error. These numbers explain why the stationary result is not used to certify the implemented continuous-time neural algorithm. Proposition 9 instead sums time-local defects over the actual finite horizon. The 80 exact-rational tests enumerate small finite Bellman problems with policy-specific evaluation and greedy defects; they are regression tests of that statement, not tests of an unproved discretization transfer.

10. COMPLETE CONTROL-RANGE DIAGNOSTICS

The following are full-domain enclosures of the final accessible mathematical actor, including the declared output-neighborhood convention. A control-range bound does not identify the optimal control or establish that a change improves payoff.

11. NONLINEAR CONTROL: ALL DIMENSIONS AND RETAINED BUDGETS

The nonlinear experiment has three dimensions, three neural seeds per dimension, two fixed query vectors per network, and five correction budgets. Its 90 neural query objects are distinct from the 15 analytic state-uniform bounds. Each dimension also has ten zero-start-gradient query objects, ten accelerated-gradient query objects, and two L-BFGS query objects, for 66 classical query objects overall. No two-query arithmetic certificate is called a uniform cube certificate.

TABLE XI
ORIGINAL NEURAL CONTROL RANGES AND CHANGE BOUNDS

Seed	Consumption	Adjustment	Portfolio	Δc	$\Delta\theta$	Δp
17100	[0.5310, 0.7494]	[0.0940, 0.1133]	[0, 0.0874]	0.2187	0.0192	0.1007
17101	[0.5701, 0.7781]	[0.0564, 0.1372]	[-0.0063, 0.1459]	0.1859	0.0538	0.1807
17102	[0.5248, 0.7726]	[0.0844, 0.1196]	[0, 0.0948]	0.2258	0.0222	0.1341
17103	[0.5823, 0.7835]	[0.0613, 0.1315]	[-0.0063, 0.1617]	0.1742	0.0461	0.1928
17104	[0.5237, 0.7292]	[0.0935, 0.1097]	[0, 0.0883]	0.2283	0.0180	0.1126
17105	[0.5754, 0.7749]	[0.0811, 0.1233]	[-0.0015, 0.1457]	0.1788	0.0286	0.1748
17106	[0.5233, 0.7479]	[0.0889, 0.1189]	[0, 0.1019]	0.2308	0.0224	0.1202
17107	[0.5819, 0.7819]	[0.0412, 0.1208]	[-0.0074, 0.1351]	0.1733	0.0643	0.1739
17108	[0.5364, 0.7447]	[0.0924, 0.1194]	[0, 0.0912]	0.2165	0.0245	0.1073
17109	[0.5864, 0.7688]	[0.0792, 0.1362]	[-0.0002, 0.1523]	0.1685	0.0440	0.1865

TABLE XII
NONLINEAR STORED-PLAN ACCURACY AT ALL RETAINED BUDGETS

Dimension	Corrections	Neural + gradient	Zero + gradient	Accelerated
8	0	1.161	9.03	9.03
8	8	1.652×10^{-5}	8.292×10^{-5}	6.636×10^{-5}
8	16	5.848×10^{-9}	1.485×10^{-8}	5.665×10^{-9}
8	24	3.18×10^{-12}	3.837×10^{-12}	2.795×10^{-13}
8	32	2.089×10^{-15}	1.612×10^{-15}	1.544×10^{-17}
32	0	3.973	36.29	36.29
32	8	5.7×10^{-5}	0.000345	0.0002687
32	16	1.505×10^{-8}	6.218×10^{-8}	2.346×10^{-8}
32	24	5.932×10^{-12}	1.607×10^{-11}	1.143×10^{-12}
32	32	2.917×10^{-15}	5.997×10^{-15}	6.451×10^{-17}
128	0	39.74	144.6	144.6
128	8	0.0004278	0.001383	0.001075
128	16	8.924×10^{-8}	2.487×10^{-7}	9.381×10^{-8}
128	24	2.92×10^{-11}	6.433×10^{-11}	4.567×10^{-12}
128	32	1.626×10^{-14}	2.057×10^{-14}	2.6×10^{-16}

Query-specific total-cost bounds, not per-coordinate losses and not cube-uniform bounds.

For each dimension and correction budget, the next table gives the largest directed stored-plan loss over all retained query objects of each method. The neural maximum is over all six seed/query objects; the other maxima are over the same two initial query vectors. L-BFGS is reported separately because its retained stopping iteration differs from the fixed correction budgets.

Every query certificate includes its objective interval and a rigorously evaluated gradient norm. When the resulting optimal-cost lower bound is positive, the record also reports relative excess cost; otherwise the relative field is absent rather than normalized by an arbitrary positive constant. The complete timing record separates one-time neural training, accumulated online correction time, and independent validation of each retained plan. A retained certificate near machine precision does not prove floating-point convergence at all states.

The nonlinear global proof uses neither a training-set generalization argument nor the two-mode Riccati identity. It bounds the full control-plan Hessian by $4I$ and $(177/8)I$, then contracts the gradient using the exact rational step $16/209$. Its operation count is conditional on the stated

strongly convex deterministic model. A stochastic nonlinear model without that property would require a different argument.

12. WELFARE AND ARITHMETIC QUALIFICATIONS

The wealth-compensation certificate checks all 20 fresh price-library actors, their original hashes, the exact original consumption cap, and both wealth faces. The added initial wealth is $39/5000$, and the constant consumption increment exactly exhausts that addition by the horizon under the original riskless drift. Preference adjustment, portfolio, price, and stopping settlement remain unchanged. The proof gives sufficient compensation for the original optimal value with an explicitly changed feasible policy, not the minimum wealth equivalent of an unchanged policy and not a neural welfare claim.

The full historical 18-node MPFR comparison and the fresh 20-node library are distinct records. Independent arithmetic cannot validate shared analytic premises merely by numerical agreement. The original rational quadrature, normal-tail, stopping, and price-envelope arguments therefore remain in the retained supplement below. The exact-real neural interpretation and the requirement that a portfolio perturbation precede the distance factor likewise remain part of the mathematical contract.

13. ORIGINAL-ECONOMY CLASSICAL AND PRICE-LIBRARY CONTROL RANGES

The certified classical control is the original-action clipping of the exact-real bilinear state interpolant of the stored tensor, held constant within a time step. At the upper wealth column its portfolio is exactly zero. Clipping corrects only the action admissibility convention; it does not multiply interior portfolios by a wealth-distance factor. A bilinear interpolant is a convex combination of four nodal values. Taking their extrema over all time slices and applying monotone action clipping therefore gives a whole-domain enclosure. The new range audit verifies all six tensors and their upper-face portfolios.

The fresh-library selector chooses one of twenty completely enumerated actors. Taking extrema over every stored consumption and preference-adjustment slab therefore also bounds every policy the selector returns for any price in $[.5, 8]$. This statement concerns the chosen policy’s controls, not the unknown optimal controls.

TABLE XIII
CONTROL RANGES IN THE UNCHANGED STOCHASTIC ECONOMY

Method	Nodes per axis	Consumption	Adjustment	Portfolio
Markov chain	9	[.425, .8]	[−.2, .2]	[0, .8]
Markov chain	17	[.2375, .8]	[−.2, .2]	[0, .8]
Markov chain	25	[.175, .8]	[−.2, .2]	[0, .8]
Semi-Lagrangian	9	[.425, .8]	[−.2, .2]	[0, .3666667]
Semi-Lagrangian	17	[.2375, .8]	[−.2, .2]	[0, .8]
Semi-Lagrangian	25	[.175, .8]	[−.2, .2]	[0, .8]
Fresh library	20 prices	[.7618225, .7721747]	[.0031022, .2]	{0}

Classical ranges hold on the complete state–time cylinder. Library ranges hold for all selected prices and all times before the original stopping time. Decimal endpoints are widened where needed. The JSON record contains the exact stored endpoints and source SHA-256 hashes.

14. PRESERVED TECHNICAL DEVELOPMENT FROM THE PRECEDING REVISION

The following R16 and R14 sections retain their historical scientific content. Statements about their experiment counts and timings describe those revisions, not additional R18 runs. Their existing source and results remain unchanged in the repository. The current results and claims are identified by the R18 entry point and the new sections above.

15. R16 EXPERIMENT IDENTITY AND PROOF DEPENDENCIES

The canonical manuscript is ECTA_R16; this is its technical supplement. The immutable input review is the R14 report dated September 22, 2026. R16 retains the original nonlinear economy, author attribution, preceding proofs, comparison data, unsuccessful experiments, and technical supplements. The new inventory model is explicitly a separate application. The repository’s R16_REVIEW.md identifies the current manuscript, response, exact source/result hashes, and canonical build evidence. A staged source transport or a branch name alone is not the reviewed scientific object.

Every numerical bound is attached to a stored mathematical object. The original neural policies use the accessibility-preserving architecture in the main paper. The time-control policies use deterministic consumption and preference actions with zero portfolio until the original

stopping time. The inventory policies use trained time-dependent gains in a coupled linear diffusion. The MPFR audit begins from frozen historical policy and dual inputs; the fresh price experiment does not. These distinctions are maintained even when two experiments reach numerically similar utility tolerances.

15.1. *Original-economy training and complete certification*

Seeds 16100–16109 and both checkpoints are retained. Widths alternate between 16 and 32; both have two hidden tanh layers. Every actor update follows three critic residual updates. Critic and actor learning rates are .0005 and .0002, with 128 samples, double precision, and one Torch thread. The random sampling mixture includes a 25 percent boundary-layer component. The optimizer is a proposal procedure and has no certified convergence claim. The main paper reports the complete twenty-checkpoint table, including all target misses.

The MPFR certifier evaluates every interior box. There are no skipped or failed cells in the retained valid records. The four state boundaries and terminal trace are handled by the algebraic identities and inaccessibility proof, not by omitting an inconvenient face. At upper wealth the witness dominates settlement for the original-control upper comparison. Only the candidate policy’s probability of hitting that face is zero. For a head perturbation, consumption and preference remain clipped to their bounds and the portfolio perturbation precedes the wealth-distance multiplication. An additive portfolio error after that multiplication is outside the certified contract.

A neural-jet diagnostic compares the independently coded value, time derivative, state gradient, and three state Hessian components with automatic differentiation at 24 test states, and checks 72 actor components. It is an implementation diagnostic; the whole-cell MPFR enclosures and the analytic boundary proof establish the certificate. The JSON records retain, separately, positive optimal residuals, negative policy residuals, unsigned evaluation bounds, action-gap allowances, and time-slab integration. They do not identify the unknown true residual maximum or the exact amount of interval wrapping.

TABLE XIV
FIXED-OBJECT COVER REFINEMENT AND SEPARATE ARCHITECTURE ABLATION

Object	Updates	Cells	Signed budget	Unsigned budget	Seconds
Accessible, coarse	2400	1024	54.004046	84.409078	14.16
Accessible, refined	2400	8192	32.157321	55.142057	37.43
All-face ablation	800	1024	28.195472	52.129689	4.75

Signed means the integrated positive optimal residual plus negative policy residual. Unsigned is the separately recorded $2e + q$ budget. Every trace budget is zero. The ablation is a different 800-update network, not a cover refinement of the 2,400-update candidate. Differences between enclosures are not certified estimates of the exact policy loss.

The all-face hard-trace network has the same invariant actor, seed, width, and 800-update budget specified in the ablation. It is not substituted for an unfavorable primary checkpoint. A smaller finite-resolution bound in that ablation would not refute the proved trace discontinuity: the incompatibility concerns a sequence of continuous all-face witnesses with vanishing global positive residual. Conversely, eliminating the trace budget does not prove that the resulting optimizer attains a small residual.

15.2. *Two same-model state-space methods*

The controlled Markov-chain generator uses upwind drift and two wide-jump diffusion directions with positive weights. The correlation enters those directions, and the stay probability is checked. The semi-Lagrangian generator uses a four-point weak Euler shock rule and bilinear state interpolation. Both apply the original stopping payoff when an approximate jump first leaves the domain, using the straight segment to locate that crossing. Neither treats this discrete boundary approximation as an exact continuous-time stopping simulation.

The three refinements use 9, 17, and 25 state nodes per coordinate and 64, 128, and 256 time steps. Their consumption, preference, and portfolio-head counts are (3, 3, 4), (5, 5, 6), and (7, 7, 8). Bounded bilinear state heads, held piecewise constant in time, define the transferred policies. These policies use the same invariant portfolio-distance factor as the neural class. The separate original-economy verifier encloses every interpolation cell and time-piece range against the continuous control maximum. Its common critic is fixed by the protocol, not selected after seeing the baseline outcomes. Discrete central values are stored as diagnostics and are never used as verified policy values.

The two implementations provide distinct state-space candidate generators. They are not claimed to be optimally tuned implementations or a complete comparison against the strongest possible classical approaches. The common critic is a limitation of the current matched verification exercise; all six resulting bounds miss .01. The paper therefore reports generation and verification costs without claiming a matched-accuracy winner.

15.3. *Established arithmetic and its remaining trusted components*

The R16 arithmetic kernel calls MPFR with 128-bit precision. Lower and upper endpoints use round-toward-minus-infinity and round-toward-plus-infinity for basic operations and elementary functions, including conversion to binary64. Scalar products and reductions propagate intervals instead of treating round-to-nearest BLAS output as exact. Rational constants are enclosed before use. The library rejects invalid logarithms, square roots, nonfinite endpoints, and division through zero. The remote build uses the installed MPFR headers; the source also contains a documented LP64 ABI fallback for the local development environment.

The complete eighteen-node audit retains every primal value interval, upper dual bound, expenditure interval, source-box count, and exact envelope breakpoint. Its C arithmetic and elementary functions differ materially from the retained rational-Taylor kernel. Its analytic conjugate decomposition, rational Gauss roots, derivative remainder argument, stopping corrections, and price-envelope mathematics are shared. Agreement therefore reduces arithmetic uncertainty but does not independently prove those mathematical premises. The 450 rational arithmetic checks, the neural automatic-differentiation comparisons, and the historical high-precision point comparisons are diagnostics, not formal interval proofs.

15.4. *Fresh model-to-price-envelope reproduction*

The fresh calculation initializes only the two endpoint prices. At every generated node, deterministic initializations produce a new feasible 16-slab policy and dual pilot; exact-decimal model constants are unchanged. The full original stopped payoff and adapted-control upper comparison are re-evaluated. The next node is the worst current exact envelope breakpoint rounded to spacing $1/1024$; a duplicate invokes the widest-interval midpoint. The declared termination rule is a proved uniform gap below .01 or thirty generated nodes. All intermediate node additions, optimizer statuses, costs, and certificates are retained, not only the final successful library.

TABLE XV
COMPLETE MPFR RE-EVALUATION OF ALL EIGHTEEN HISTORICAL PRICE NODES

Price	Policy lower	Optimal upper	Node gap
0.5	-1.2726630012	-1.2628181763	0.0098448248
0.875	-1.2789733032	-1.2692555100	0.0097177931
1.25	-1.2847932336	-1.2751091803	0.0096840532
1.625	-1.2901339789	-1.2804567231	0.0096772557
2	-1.2950178727	-1.2853301725	0.0096877002
2.5	-1.3008396015	-1.2911285459	0.0097110555
3	-1.3059053778	-1.2961649251	0.0097404526
3.5	-1.3102486540	-1.3004746771	0.0097739768
4	-1.3138977523	-1.3040895090	0.0098082433
4.25	-1.3154696749	-1.3056468359	0.0098228390
4.5	-1.3168890699	-1.3070533282	0.0098357416
5	-1.3193518240	-1.3094933092	0.0098585147
5.5	-1.3214155553	-1.3115375893	0.0098779659
6	-1.3231704727	-1.3132756877	0.0098947850
6.5	-1.3246813844	-1.3147719178	0.0099094666
7	-1.3259960813	-1.3160736824	0.0099223988
7.5	-1.3271505957	-1.3172167379	0.0099338578
8	-1.3281726192	-1.3182285110	0.0099441081

Lower endpoints are rounded downward and upper bounds upward. Differences are computed from exact dyadic endpoint fractions, not from the displayed decimals. Every original adapted control is covered by the upper bound; every lower bound belongs to its identified stored time-control policy. The exact all-price envelope is computed from unrounded endpoints and expenditure intervals.

The canonical result has 20 newly generated nodes. It uses the rational–Taylor arithmetic inherited as an algorithm, not historical policy data or preselected dual pilots. The separate historical MPFR result must not be reassigned to this fresh object. Neither price-node refinement nor a denser quadrature eliminates the current dual relaxation’s positive allowances; the retained 16/32/64-slab experiment records the corresponding floor.

15.5. *Coupled-state neural control*

The inventory model, neural parameterization, Riccati residual, and global proof are given in the main paper. Seeds 16200–16209 alternate between widths 8 and 16, with two hidden tanh layers. Adam uses learning rate .005, batch 128, double precision, and one thread. Both 250- and 1,000-update checkpoints are retained. Each is certified on 512 complete time cells. One pair of gain functions is reused at dimensions 4, 8, 16, 32, 64, and 128. The proof covers unbounded state dynamics and every initial state with mean squared inventory at most one; no simulated tail correction or finite state cover is used.

The resource record also includes a structure-aware classical solution of the two scalar Riccati equations using adaptive floating-point RK45. This is a diagnostic reference and is not an interval certificate, a neural training label, or a claimed matched-verified-accuracy benchmark. The theorem compares with the exact Riccati optimum through a differential error inequality, not by treating that numerical reference as ground truth. The per-coordinate and total bounds are both stored; multiplying by dimension is not evidence of dimension-independent total welfare error.

TABLE XVI
FIXED NEURAL GAIN FUNCTIONS: COMPLETE TIME-COVER REFINEMENT

Time cells	Orthogonal residual	Mean residual	Loss/coordinate
128	0.02124605	0.03542657	0.000411639
512	0.01378842	0.01707312	0.000095606
2048	0.01193411	0.01477912	0.000071640

The same seed 16200 final gain pair is used in every row. This isolates enclosure refinement from training-budget changes. All state dimensions share the same two-mode proof; total loss is dimension times the per-coordinate bound.

TABLE XVII
INVENTORY TOTAL POLICY-LOSS BOUNDS ACROSS STATE DIMENSION

States	Smallest final bound	Largest final bound
4	0.00010119	0.00038243
8	0.00020237	0.00076485
16	0.00040473	0.00152969
32	0.00080945	0.00305938
64	0.00161889	0.00611876
128	0.00323777	0.01223752

The range is over all ten prescribed final checkpoints, not a selected training result. These are total loss bounds. The initial region scales as $|x_0|^2/d \leq 1$; the full diffusion is state-global. Every dimension reuses the same gain functions and invariant-mode proof, so these rows are not six independent training/verification cost measurements.

15.6. *Economic interpretation and reproducibility*

The consumption-top-up calculation keeps the preference, adjustment-cost, wealth, and stopping paths fixed while providing externally financed consumption goods. It does not assert budget-feasible wealth compensation or empirical calibration. The main paper gives the marginal-utility inequality, discounted alive-time lower bound, and scale qualification. The original strict-loss and relative-resolution counterfactual diagnostics are preserved below. Policy-control accuracy is not inferred solely from a small value-loss bound; the selected time-control ranges remain distinct from the unknown optimal feedback.

TABLE XVIII
MEASURED EXPERIMENT RESOURCES

Experiment	Wall seconds	Reported peak RSS (MiB)
Original neural	1093.74	—
Inventory neural	75.98	333.73
Fresh price library	415.79	497.42
MPFR historical nodes	1065.89	276.26
Six classical candidates	95.85	—

Per-experiment timings exclude package installation. Neural seeds and arithmetic nodes may run concurrently; peaks are process-local where stated in JSON and must not be summed as simultaneous memory. A dash means no aggregate peak was measured; seed/process peaks remain in their individual records. The root pipeline receipt reports end-to-end wall time and the execution environment identifies hardware and package versions.

From a clean checkout, the command

`python revisions/2026-09-23-r16/replication/run_all.py`
 executes the new scientific pipeline. Existing experiment output directories are refused rather than overwritten. The run records the source commit, source hashes, package versions, processor/platform information, per-process resources, and every attempt. The paper table generator reads the resulting JSON and rounds all bound decimals upward. The publication validator compiles both documents, checks internal references, verifies complete counts and target labels, and hashes the materialized scientific objects. The public review receipt identifies the scientific source and result commits separately from any subsequent publication metadata.

The following sections preserve the R14 supplement in full. They describe the retained R14 implementation and historical objects, except where an R16 replacement has been explicitly identified above. Their existence does not make those historical neural policies newly successful or convert a diagnostic comparison into a rigorous proof.

16. RETAINED R14 TECHNICAL DEVELOPMENT

17. SCOPE, NOTATION, AND THE CURRENT EVIDENCE

This supplement supplies details required to verify the R14 manuscript. Time, domain, control bounds, utility, and first-exit settlement are exactly those in the main paper. In particular, the lower calculations do not replace stopped CRR utility by an unstopped singular expectation, and the upper calculations do not restrict the original adapted portfolio control to zero. The deterministic policies are lower-bound objects; their upper comparison bounds optimize over the original economic opportunities.

Three numerical objects have distinct meanings. The state-feedback tanh actor/critic has a full-domain residual certificate whose current bound is large. The independently verified time-control library has a sharp central-state price-continuum certificate and a new initial-state rectangle certificate. The freshly generated 16-, 32-, and 64-interval policies provide a policy-richness and resource experiment without inherited optimizer inputs. These objects have separate files, hashes, and accuracy predicates.

The main paper contains full proofs of its residual theorem, coefficient transport, and four-corner price construction. Sections below give the analytic economic dual and the independently implemented enclosure lemmas used as inputs to the computer-assisted economic propositions. The shared analytic premise is explicit: two numerical implementations of a false analytic inequality would not establish that inequality.

18. THE AFFINE-PREFERENCE DUAL IN THE ORIGINAL STOPPED ECONOMY

18.1. *Consumption conjugacy and its supporting plane*

Write $r = u - 1$, $l = \log y$. On the rectangle relevant to the witness, the lower consumption cap .05 never binds. The consumption conjugate is

$$\Psi(u, e^l) = \begin{cases} -\exp(-r \log .8)/r - .8e^l, & l \leq -u \log .8, \\ -u \exp(rl/u)/r, & l \geq -u \log .8. \end{cases} \quad (1)$$

Let $S_0 = \Psi + .01e^l - .004 \log(.5)$ and $\beta = \partial_u \Psi$. Both branches match in value and first derivatives at the consumption switch. Their one-sided Hessians need not match; all interval boxes crossing the switch use their hull.

The desired supporting inequality is

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad u \in [1.5, 2.5], \quad m \in [2, 2.2], \quad y \in [.2, 12]. \quad (2)$$

The following derivative facts reduce its proof to one-dimensional covers. On an interior consumption branch, putting $L = l/u$ and $E = \exp((1 - 1/u)l)$, direct differentiation gives

$$\partial_l S_{0,uu} = \frac{EL(2-L)}{u^2}, \quad \partial_l \beta = -\frac{El}{u^2}.$$

The first derivative is nonnegative on the present box: $0 < L < 2$ whenever the interior branch applies. On the capped branch $S_{0,uu}$ is independent of l . Thus its maximum over the y rectangle is controlled at $y = 12$.

A complete interval cover of $u \in [1.5, 2.3]$ at $y = 12$, consisting of 1,600 intervals, gives $S_{0,uu} \leq -.0671742195$. Hence β is decreasing over this interval. On $u \in [2.3, 2.5]$, a separate 2,000-interval cover gives

$$\beta(u, 12) \leq -.9675376119 < -.9572021815 \leq \beta(2.2, 12).$$

To extend this comparison to smaller y , compare $\beta(u, e^l) - \beta(m, e^l)$ for $u > m$. In the interior branch, E/u^2 decreases in u because $\partial_u \log(E/u^2) = l/u^2 - 2/u < 0$ on the box. Therefore $\partial_l \beta(u, e^l) - \partial_l \beta(m, e^l) \geq 0$ whenever both are interior. When the larger u is capped and the smaller is interior, this difference is again nonnegative; when both are capped it is zero. The difference is consequently largest at $y = 12$. Integrating the resulting derivative order from m to u proves the upper-side supporting inequality. Integrating the decreasing derivative on the lower-side interval proves the other side. This proves (2) without asserting global concavity of S_0 .

Additional complete covers of $m \in [2, 2.2]$ verify

$$-7.1 \leq S_0(m, y) \leq 0, \quad |\beta(m, y)| \leq 1.1 \quad (y \in [.2, 12]). \quad (3)$$

Here S_0 is decreasing in y , since its derivative is $-c^* + .01 < 0$, and β is nonincreasing in l where its derivative is nonzero. Checking the two y endpoints therefore suffices. The recorded lower bound is above -6.989553453 , and the recorded absolute coefficient bound is below $.971182123$. All finite covers use the new interval implementation and the formulas (1) with automatic differentiation.

18.2. The auxiliary adapted-control problem

Let Y follow $dY = .02Y dt - .3Y dZ^x$, let $\hat{Y}$ be its clipping to $[\cdot 2, 12]$, and fix a deterministic reference drift $\bar{\theta} \in [0, \cdot 2]$. Put

$$m(t) = 2 + \int_0^t \bar{\theta}_s ds, \quad \phi(u) = -.02(u - 2)^2,$$

$$Q(u) = \phi(m(1)) + \phi'(m(1))(u - m(1)), \quad b_T = \phi'(m(1)).$$

Define the affine source

$$\hat{S}_t(u, y) = S_0(m(t), \hat{y}) + \beta(m(t), \hat{y})(u - m(t))$$

and the full-line auxiliary value

$$A_k(t, u, y) = \sup_{|\theta| \leq .2} \mathbb{E}_{t,u,y} \left[\int_t^1 e^{-\rho(s-t)} \{ \hat{S}_s(U_s, Y_s) - k\theta_s^2/2 \} ds + e^{-\rho(1-t)} Q(U_1) \right], \quad (4)$$

where $dU_s = \theta_s ds + .05dZ_s^u$. The preference action is still adapted. Because Y is exogenous and the coefficients of U and its terminal value are affine, the conditional marginal value is

$$B_s = \mathbb{E} \left[\int_s^1 e^{-\rho(t-s)} \beta(m(t), \hat{Y}_t) dt + e^{-\rho(1-s)} b_T \middle| \mathcal{F}_s \right].$$

Conditional Fubini, followed by pointwise maximization of the strictly concave quadratic in θ_s , gives the exact improvement formula relative to $\bar{\theta}$:

$$A_k(0, 2, y_0) = \bar{A} + \int_0^1 e^{-\rho s} \mathbb{E} g_k(B_s, \bar{\theta}_s) ds, \quad (5)$$

where $g_k(q, a) = H_k(q) - qa + ka^2/2$ and

$$H_k(q) = \max_{|a| \leq .2} \{qa - ka^2/2\}.$$

The reference expectation is

$$\bar{A} = e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \{\mathbb{E} S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2\} dt + C, \quad (6)$$

$$C = .00375 \int_0^1 t e^{-\rho t} \mathbb{E} [\partial_t \beta(m(t), \hat{Y}_t)] dt. \quad (7)$$

To obtain the covariance term, use the reference decomposition $U_t = m(t) + .05Z_t^u$. The log-deflator is $l_t = \log y_0 - .025t - .3Z_t^x$. Gaussian integration by parts gives $\mathbb{E}[Z_t^u \beta(l_t)] = .075t \mathbb{E} \beta_l(l_t)$, because the correlation is $-.25$. Multiplication by $.05$ produces $.00375$. Clipping is included in the almost-everywhere derivative. Since $\beta_l \leq 0$, the correction is nonpositive; it is evaluated rather than dropped.

18.3. A quantitative allowance for the random marginal value

The derivative H'_k is $1/k$ -Lipschitz, including across its saturation points. For $b(s) = \mathbb{E} B_s$ this yields

$$\mathbb{E} g_k(B_s, a) \leq g_k(b(s), a) + \frac{\text{Var}(B_s)}{2k}.$$

A Gaussian Poincaré inequality for the past Brownian coordinate and conditional Jensen give

$$\text{Var}(B_s) \leq .09s \left\{ \int_s^1 e^{-\rho(t-s)} \sqrt{\mathbb{E} |\partial_t \beta(m(t), \hat{Y}_t)|^2} dt \right\}^2.$$

For $m \in [2, 2.2]$, the nonzero derivative has $l > 0$ and

$$|\partial_l \beta|^2 \leq \frac{1}{16} e^{(12/11)l} l^2.$$

The derivative vanishes in the capped-consumption and clipped-deflator regions. For a Gaussian l with mean μ and variance v ,

$$\mathbb{E}[e^{al} l^2] = e^{a\mu + a^2 v/2} \{(\mu + av)^2 + v\}.$$

Using $a = 12/11$, $M = \log y_0$, $\mu \leq M$, $v \leq .09$, and the positive mean range $y_0 \in [1.2, 2.4]$, define

$$D^2 = \frac{e^{aM + .045a^2}}{16} \{(M + .09a)^2 + .09\}. \quad (8)$$

The preceding expressions are increasing in the relevant μ, v ranges. Since $\int_0^1 s(1-s)^2 ds = 1/12$ and $C_\rho(1-s) \leq 1-s$, equations (5)–(8) imply

$$A_k(0, 2, y_0) \leq \bar{A} + \int_0^1 e^{-\rho s} g_k(b(s), \bar{\theta}_s) ds + \frac{.09D^2}{24k}. \quad (9)$$

This positive variance allowance is retained at every mesh. An unconditional reference control is not silently treated as the optimal adapted auxiliary control.

18.4. Returning to the first-exit contract

Set $h = 1 - t$ and

$$W(t, u, x, y) = y(x - .5) + .1 \log(.5) + A_k(t, u, y).$$

The product identity for Yx cancels every adapted portfolio control. Consumption conjugacy and the supporting plane imply that discounted W plus the original running payoff is a supermartingale up to the relevant state or localization face. This can be proved directly by conditional Fubini and the adapted selector in (5). Alternatively, smoothing the bounded Lipschitz source with a uniform error correction gives a classical verification argument and then the limit.

The affine tangent satisfies $Q \geq \phi$. To control continuation, use the feasible auxiliary action $\theta = 0$. For an initial $u \in [1.5, 2.5]$,

$$\mathbb{E}|U_r - m(r)| \leq .7 + .05 = .75, \quad \mathbb{E}|Q(U_r)| \leq .0008 + .008(.75) = .0068.$$

Applying Itô to Q and using (3) shows

$$A_k(t, u, y) - Q(u) \geq \{-7.1 - 1.1(.75) - .04(.0068)\}h = -7.925272h > -8h. \quad (10)$$

For $x \in [.5, 2]$, $y \geq .2$, $y(x - .5) + .1 \log(.5) - .1 \log x \geq 0$, by differentiation in x and equality at $.5$. Thus W dominates the original settlement on actual wealth exits and at the horizon.

For completeness, G is a global upper comparison for the original economy. The utility is increasing in consumption, and even its largest value on the original state/action box is less than $-.82$. The positive parts of θG_u , the wealth drift term, and $-\rho G$ together are less than $.017$, while the remaining diffusion and adjustment terms are nonpositive or can be bounded within this margin. Hence $\sup_a \mathcal{R}_a G < 0$. Also $G \geq g$ on the stopping trace. Stopped verification therefore gives $V_k \leq G$ for $k \geq .5$.

The potentials

$$Z_u = 8h \{e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h}\},$$

$$Z_y = 8h \{e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h}\}$$

are nonnegative supermartingale potentials. Indeed, $.2(120) + .00125(120)^2 = 42$ bounds the preference exponential generator, while $.025(22) + .045(22)^2 = 22.33$ bounds either sign of the log-deflator generator. Differentiating the factor h makes the killed generator nonpositive.

Each contributes at least $8h$ at its artificial face. Adding them to W , using (10) and then the fallback $V \leq G$, proves the original-economy upper bound in the main paper. Artificial clipping and full-line continuation are therefore proof devices with explicit allowances, not changes to the economic contract.

In the stress calculation the corresponding continuation floor is $-7.11 - 1.11(.75) - .04(.0068) = -7.942772 > -8$. Witness perturbations remain in the same verified reference class, so the same analytic argument applies.

19. INDEPENDENT ARITHMETIC AND QUADRATURE

19.1. *Trusted arithmetic model*

The interval core assumes IEEE 754 binary64 addition, subtraction, multiplication, and division in round-to-nearest arithmetic; correctly directed neighboring numbers from `nextafter`; exact binary decomposition/scaling through `frexp` and `ldexp` where applicable; and gradual underflow rather than flush-to-zero. Each elementary result is expanded outward. Reductions use explicit outward pairwise operations. Neural affine layers do not invoke an unaccounted BLAS dot-product error model. Overflow, nonfinite endpoints, or division through zero cause failure rather than cell deletion.

Every decimal economic constant is first interpreted as a rational. Every stored binary64 parameter is its exact dyadic rational value. A rational is converted to an interval only after checking its position against the candidate floating value. Intervals can overestimate because of dependency, but cannot be narrowed by an optimizer's successful termination flag.

For $|x| \leq 1/8$, the exponential uses its degree-18 Taylor polynomial with the uniform absolute remainder

$$\left| e^x - \sum_{j=0}^{18} \frac{x^j}{j!} \right| \leq \frac{8}{7} \frac{(1/8)^{19}}{19!}.$$

Larger arguments are divided by powers of two until the small-argument condition holds, and enclosed squaring restores the result. For a positive argument, an exact binary exponent and a mantissa in $[1, 2)$ reduce its logarithm to

$$\log m = 2 \sum_{j=0}^{39} \frac{y^{2j+1}}{2j+1} + R, \quad y = \frac{m-1}{m+1}, \quad |R| \leq \frac{2(1/3)^{81}}{81(1-1/9)}.$$

The same construction encloses $\log 2$. Tanh is evaluated from the enclosed exponential and monotonicity, with exact range restriction to $[-1, 1]$. No assumed ulp accuracy of a platform transcendental function enters these proof computations. Separate tests compare 807 elementary enclosures with 100-digit evaluations; these tests check implementation behavior but do not replace the remainder inequalities.

19.2. *Rationally certified Gauss rules*

The eight Gauss–Legendre roots are first proposed by an ordinary floating-point routine. Each root is then isolated in a disjoint rational bracket by exact sign evaluations of the degree-eight Legendre polynomial and 80 bisections. Eight disjoint sign-changing brackets account for all roots. The positive weights are enclosed using

$$w_i = \frac{2}{(1-x_i^2)[P'_8(x_i)]^2}.$$

The program checks positivity and all moments through degree 15. The Gauss rule’s exactness through degree $2n - 1$ and its positive weights are the standard construction described in NIST (2026); its nodes and weights are not assumed to equal their floating-point proposals.

Square-root endpoint proposals are validated by exact rational square comparisons and expanded until they bracket the true root. The constant π is enclosed using Machin’s identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with 60 alternating-series terms and the first omitted term as a remainder bound.

For an analytic function bounded by M on a complex disk of radius R around the center of a real integration interval of half-width $h < R$, the degree-15 Taylor remainder on that interval is at most $M(h/R)^{16}/(1 - h/R)$. Both integration and the positive Gauss rule have norm equal to the interval length. Subtracting their exact actions on the degree-15 polynomial gives the rigorous remainder

$$|I(f) - Q_s(f)| \leq 2 \text{length } M \frac{(h/R)^{16}}{1 - h/R}. \quad (11)$$

This inequality, interval root/weight errors, and outward summation are used together. Agreement between two mesh levels is not an error bound.

19.3. The original-policy evaluator

For a stored time policy and an initial state in K , let μ_j be its cumulative preference drift at the start of interval j . After $t = v^2$ the covered preference is

$$U = u_0 + \mu_j + \theta_j(v^2 - j/n) + .05vz, \quad z \in [-8, 8].$$

The integrand contains the original utility $-\exp[-(U - 1) \log c_j]/(U - 1)$, the Gaussian density, the Jacobian $2v$, and discount $e^{-.04v^2}$. It is not a polynomial utility surrogate. A tensor Gauss rule uses eight subdivisions in v per policy interval, 32 normal cells, and eight Gauss nodes on each axis.

The complex-neighborhood constants used in (11) are global on the state rectangle and the feasible control ranges. For the v disk use radius $1/8$ with real $z \in [-8, 8]$. One has $|v| \leq 1.125$, and the real part of $U - 1$ stays between .476875 and 1.723125. There is no CRRA pole in the disk. The utility, Gaussian density, discount, and Jacobian together have modulus less than five. For the z disk use radius two with real $v \in [0, 1]$; the real part of $U - 1$ stays between .48 and 1.72, while the Gaussian density’s complex factor is at most $e^2/\sqrt{2\pi}$. The complete modulus is below 32. These values give $M_v = 5$, $M_z = 32$. Tensor error follows by writing $I_v I_z - Q_v Q_z = (I_v - Q_v)I_z + Q_v(I_z - Q_z)$ and using positivity and the interval lengths. All remainder terms are summed outward.

To bound the omitted normal tails, extend the running utility by clipping its preference argument to $[1.2, 2.8]$ outside the covered range. On the feasible consumption interval this extension lies in $[-6, 0]$. Its omitted expected contribution lies in $[-12e^{-32}, 0]$, by the Gaussian Chernoff bound. The terminal preference quadratic is integrated analytically without clipping, and terminal wealth is exactly enclosed from the budget. The auxiliary full-horizon functional is connected to the original stopped payoff by a separate estimate; it is not substituted for that payoff without correction.

For all initial states in K and all stored or fresh feasible drifts in $[0, .2]$, either preference exit requires a Brownian excursion of at least .58 after subtracting the largest possible drift. Therefore

$$p_{\text{exit}} \leq 2 \exp\{-.58^2/(2 \cdot .05^2)\}.$$

Also

$$\mathbb{E}(U_1 - 2)^4 \leq .22^4 + 6(.22)^2(.0025) + 3(.0025)^2.$$

Bounding the running-payoff, fee, and bounded terminal differences on the exit event, and using Cauchy–Schwarz for the unbounded terminal quadratic, gives the uniform allowance

$$\epsilon_{\text{stop}} = 15p_{\text{exit}} + .02\sqrt{\{.22^4 + 6(.22)^2(.0025) + 3(.0025)^2\}p_{\text{exit}}} < 3.87 \times 10^{-18}. \quad (12)$$

The constant 15 covers all $k \leq 8$. The wealth path is strictly decreasing for the verified controls, with $x_1 > .5$, so there is no wealth exit before time one. The final policy interval includes the symmetric quadrature error, the one-sided missing utility tail, and the symmetric stopping correction (12).

The central-state stopped expenditure is independently enclosed by its exact deterministic full-horizon value minus at most $.02p_{\text{exit}}$. The state–price concavity construction instead uses the full-horizon expenditure of its covered lower functional; its uniform stopping error has already been paid. Both constructions are therefore valid, but their expenditure labels must not be interchanged.

19.4. *The independent dual evaluator*

The dual changes variables to $t = v^2$, with 32 subdivisions per reference time interval and 512 normal cells on $[-6, 6]$. The source and its first and second derivatives are generated from (1) by an independent two-variable interval differentiation class. Chain rules map them to (v, z) . Every source box is enclosed, and every possible consumption-switch branch is included. The program verifies that the normal cover does not cross a deflator clipping face; the omitted tails cover the remaining probability.

Write $w_v = 2ve^{-.04v^2}$ and $w_z = (2\pi)^{-1/2}e^{-z^2/2}$. Zeroth, first-centered, and second-centered moments of these weights are integrated by the validated Gauss rule. For their Cauchy bounds the disks have radii $1/8$ and two and base modulus bounds three. Centered moment factors use $(R + 1/1024)^p$, $p = 0, 1, 2$, with an explicit check that midpoint representation errors are below $1/1024$. This accounts for a floating-point center that is not an exact interval midpoint.

Constant and linear source terms retain the signed first-centered moments. The source Hessian on the complete box bounds the remainder by

$$\frac{1}{2}\|S_{vv}\|_{\infty}M_{v,2}M_{z,0} + \|S_{vz}\|_{\infty}r_vr_zM_{v,0}M_{z,0} + \frac{1}{2}\|S_{zz}\|_{\infty}M_{v,0}M_{z,2}.$$

The source is C^1 across the action switch with bounded one-sided Hessians, so the integral Taylor formula with the Hessian hull is valid there. Weighted first moments are not discarded on the false premise that a weighted box is symmetric.

The Gaussian tail probability is at most $2e^{-18}$. The source tail is enclosed between -7.1 times its discounted mass and zero; the coefficient tail has absolute bound 1.1 times its mass. Since $\beta_i \leq 0$ and $|\beta_i| \leq 2.5$ on its nonzero clipped region, the covariance tail has the signed enclosure used in the code. These corrections cover the entire normal distribution.

To enclose $b(s)$, coefficient integrals are accumulated backward in time. The incomplete current cell is bounded by its entire discounted mass and its coefficient interval. The convex function $g_k(q, \bar{\theta})$ attains its maximum on the resulting q interval at an endpoint. Its scalar action maximum is evaluated with the clipped maximizer. Terminal, source, covariance, control-gap, variance, localization, and arithmetic terms are finally summed outward. The stored upper endpoint is a bound on the original optimum, not on the fitted pilot objective.

20. INDEPENDENT HIGH-PRECISION CROSS-CHECKS

A third program evaluates the Gaussian expectations by closed truncated-lognormal moments rather than the production or second interval quadrature. If $L \sim N(\mu, s^2)$, define

$$M_q(a, b) = e^{q\mu + q^2 s^2 / 2} \left\{ \Phi\left(\frac{b - \mu - qs^2}{s}\right) - \Phi\left(\frac{a - \mu - qs^2}{s}\right) \right\}.$$

The first weighted moment is

$$M_{1,q}(a, b) = (\mu + qs^2)M_q(a, b) + se^{q\mu + q^2 s^2 / 2} \{\varphi(z_a) - \varphi(z_b)\}.$$

The capped region, interior consumption region, and two deflator-clipping tails are integrated separately using these formulas. In particular, the interior coefficient derivative integrates as $-M_{1,q}/m^2$, not by blindly applying a smooth quadrature rule across its switch. Remaining time integrals are evaluated at 40 and 70 decimal digits.

The source, initial affine coefficient, and covariance at prices .5, 4.25, and 8 all fall inside the independently certified intervals at both precisions. All six runs and their timings are stored. This test is numerically independent in decomposition and precision, but its time quadrature is not asserted to produce an interval proof. The decisive bounds remain those from the complete rational-root/interval execution.

21. NEURAL DERIVATIVES, ACTIONS, AND THE OUTPUT-ERROR CONTRACT

For a scalar composition $w = \tanh z$, the chain rule is

$$w_i = (1 - w^2)z_i, \quad w_{ij} = -2w(1 - w^2)z_i z_j + (1 - w^2)z_{ij}.$$

The interval implementation propagates value, t, u, x first derivatives and uu, ux, xx second derivatives through each affine layer and tanh layer. For the critic $v = G + (1 - t)N$, this gives, for example,

$$\begin{aligned} v_t &= -N + (1 - t)N_t, & v_{ux} &= (1 - t)N_{ux}, \\ v_{uu} &= -.04 + (1 - t)N_{uu}, & v_{xx} &= -.1/x^2 + (1 - t)N_{xx}. \end{aligned}$$

Every multiplication and layer reduction is outward. The full cell image is used, not a derivative evaluated only at its center.

The action oracle uses the scalar formulas in the main paper. If the marginal-wealth interval crosses zero, the consumption optimum is enclosed over the whole feasible interval. If the portfolio-curvature interval crosses zero, the complete portfolio range is retained in the upper evaluation. These cases can make the bound wide, but preserve validity for all continuous actions. The candidate action ranges additionally include the 2^{-24} output allowance and are clipped to the exact economic action bounds.

The residual and action records for every interior cell are stored in compressed JSON. The finest cover has 16,384 such cells; each of the four first-exit faces has 512 cells. The terminal face is exact by the critic construction. The per-time-slab maxima are assembled with an enclosed discount integral. A separate undiscounted integral over the full horizon supplies a safe all-initial-times bound. The model-transport modulus encloses drift, variance, cross-covariance, discount, and running-cost changes simultaneously; the correlation remains fixed at $-.25$.

The mathematical feedback uses exact stored weights and real tanh. Its output-neighborhood certificate covers any admissible implementation satisfying the stated action-error contract.

No automatic guarantee for a hardware tanh implementation, a stochastic time integrator, or a floating-point budget reconstruction follows from this contract without a separate error analysis. The network’s fine-domain certificate is deliberately not replaced by the much sharper bound belonging to a different time-control policy.

22. POLICY INTERPRETATION AND ECONOMIC-RESOLUTION RECORDS

TABLE XIX
CENTRAL-STATE SELECTED POLICY REGIONS AND CONTROL SUMMARIES

Policy k_j	Left	Right	Width	Consumption range	Mean θ
0.5	0.50000	0.67789	0.17789	0.76182–0.77217	0.18363
0.875	0.67789	1.06576	0.38787	0.76197–0.77151	0.17275
1.25	1.06576	1.43620	0.37044	0.76217–0.77092	0.16279
1.625	1.43620	1.81288	0.37668	0.76243–0.77039	0.15319
2	1.81288	2.24707	0.43419	0.76273–0.76995	0.14390
2.5	2.24707	2.74540	0.49833	0.76320–0.76944	0.13209
3	2.74540	3.24895	0.50355	0.76374–0.76903	0.12073
3.5	3.24895	3.74918	0.50023	0.76435–0.76872	0.10983
4	3.74918	4.12169	0.37251	0.76499–0.76849	0.09950
4.25	4.12169	4.37187	0.25017	0.76529–0.76840	0.09446
4.5	4.37187	4.73837	0.36650	0.76557–0.76833	0.08992
5	4.73837	5.23938	0.50102	0.76605–0.76822	0.08205
5.5	5.23938	5.74023	0.50085	0.76646–0.76815	0.07546
6	5.74023	6.24095	0.50072	0.76632–0.76812	0.06985
6.5	6.24095	6.74158	0.50062	0.76616–0.76811	0.06503
7	6.74158	7.24212	0.50054	0.76602–0.76811	0.06083
7.5	7.24212	7.74259	0.50047	0.76590–0.76814	0.05715
8	7.74259	8.00000	0.25741	0.76579–0.76818	0.05389

Endpoints and widths are descriptive rounded values; exact rational endpoints determine the selector. Consumption and average preference adjustment refer to the selected stored policy, with portfolio zero. The state-uniform selector is separately recorded because it minimizes the worst corner gap rather than only the central-state gap.

The exact price program records strict-loss and relative-resolution regions for four anchor prices. Weak monotonicity holds in all cases from the model. A missing strict-resolution region means that the certified interval still includes zero, not that a price increase benefits the decision maker. The relative criterion for complete regions uses the untruncated interval difference and is conservative; selected pairwise tables also use the universal expenditure bound to tighten the upper loss endpoint.

23. REPLICATION, SCOPE OF TIMING, AND HISTORICAL PRESERVATION

The replication directory contains separate entry points for neural generation, neural verification, independent policy evaluation, independent dual evaluation, exact price postprocessing, state–price certification, full-library stress, high-precision diagnostics, and the fresh classical frontier. Executed results are written to new directories rather than overwriting earlier attempts. The build and manifest checks distinguish a numerical failure to meet a requested tolerance from a malformed or missing certificate.

The neural run is a full fresh run after interpreter entry, including all critic and actor updates. Shell timing separately includes imports. The classical frontier separately measures policy generation, dual fitting, primal verification, and dual verification, starts from fixed initial guesses in every case, and retains optimizer reports. The old library audit starts from frozen inputs and is labeled verification only. Historical interrupted work not recorded in the archive is unknown, not zero. Resource clocks from overlapping runs cannot establish a matched-hardware speed ranking.

A source-frozen clean replay reexecutes all 18 second-implementation node calculations and compares the decisive lower endpoints, upper endpoints, and newly computed expenditure intervals exactly. The original development ledger is preserved, including the fact that expenditure fields were subsequently replaced by independent calculations. The clean replay checks the final source rather than silently assigning its hash to an earlier execution. Content hashes identify every included source and result file. A local reconstruction of the R12 archive is distinguished from its actual remote Git commit; publication metadata does not claim a remote write when none has occurred.

Historical preservation is handled at the file level, not by requiring every previous sentence to remain in the current main text. A byte-for-byte audit compared all 1,830 files of the downloaded R12 repository archive with the initial working copy and found no mismatch. Earlier manuscripts, proofs, data, failure records, and code remain unchanged. The old navigation index is archived before its replacement with a single R14 entry point. The current supplement supplies the proofs needed for the current manuscript; it does not concatenate 148 pages of superseded manuscripts.

TABLE XX
PRESERVATION MAP FOR SUBSTANTIVE INHERITED MATERIAL

Material	Unchanged location and interpretation
Original model and early theoretical development	Root <code>ECTA.tex</code> , <code>ECTA_R2.tex</code> , and earlier source directories; retained research history.
Recursive utility, sophisticated selves, games, stochastic trace identities	Earlier manuscripts and <code>SUPP_R12.tex/.pdf</code> ; retained extensions, not new full-domain numerical claims.
Original continuous-policy evaluation and cover/oracle theory	R9 replication and proof directories; comparison baseline for the independently implemented R14 objects.
Affine-preference dual and earlier loose witnesses	R10 proof, result, and retained-construction files; analytic history and failed/loose certificates remain visible.
External holdout and method-specific tuning	R9–R11 result and paper directories; unfavorable panels and unstable seed orderings preserved.
Exact-reference gain-policy study	R11 accuracy source and results; not recast as the R14 tanh architecture’s accuracy.
Structured 128-dimensional action oracle	R10–R12 retained action-oracle evidence; frozen-jet action optimization, not 128-state dynamic control.
Eighteen-price continuation and historical refinement	R12 source, results, receipts, and refinement history; frozen inputs and their original timing scope preserved.

The response document maps every R13 finding, the inherited R12 issues, and the current evidence paths. It explicitly distinguishes completed calculations from unachieved neural accuracy, absent matched-accuracy classical comparisons, and remote publication status. This preserves the research objective without certifying a claim that the current computation has not established.

24. HISTORICAL R18 EXPOSITION, PRESERVED VERBATIM

The next two sections reproduce the preceding revision’s introduction and conclusion without deletion. Their descriptions of the then-primary method and then-available evidence are historical. The new policy-value mechanism and new evidence are stated in the R19 main text; this annex is not an alternative current claim of numerical performance.

25. INTRODUCTION

Neural approximations to dynamic economic decisions require a criterion that is linked to the value of the policy actually delivered. Average equation residuals need not provide that link. A small error at frequently visited training states can coexist with a large error near a stopping face, and improving a value witness need not improve the policy with which it is paired. These distinctions become operational when an algorithm must return a complete error bound rather than a sampled diagnostic.

This paper develops certificate-aligned neural Bellman operators. The primary algorithm minimizes a differentiable envelope of two signed Bellman residuals on a complete state–time cover. An independent directed-arithmetic verifier checks the proposed policy and witness against the original economic model. Every prescribed proposal is retained; an incumbent is replaced only when its complete certificate decreases. A quantitative theorem connects the trainable envelope to the rigorous policy-loss bound, separating the smoothing allowance from a checkable numerical correction. The theorem concerns the loss actually used in the computation, rather than an unexecuted policy-iteration idealization.

Stopping geometry is part of this construction. In our consumption–investment economy with costly preference adjustment, continuation value has a strict jump exceeding 6.37 utility units at an inward upper-wealth face. A critic forced to equal immediate settlement on every face therefore cannot acquire a vanishing uniform positive residual. A signed accessibility-weighted comparison permits the critic to dominate settlement at a face that the candidate cannot reach, while retaining every originally admissible action in the optimal upper comparison. This corrects an approximation-space inconsistency without modifying preferences, controls, or the stopping contract.

The original-economy experiment compares two generations of training on that unchanged problem. Under the historical sampled objective, extending training from 800 to 2,400 updates worsens all ten certificates, even though sampled residual mean squared error falls for nine seeds. Under the complete-cover objective, all ten accessibility-aware trajectories have decreasing raw certificates at 0, 100, and 400 updates. Their final bounds range from 7.27832 to 7.34367, compared with initial bounds from 26.3005 to 26.6668. A paired ten-seed all-face experiment, unrestricted state-space comparators with method-specific witnesses, and a fixed-network refinement study identify what causes this improvement and what it does not establish. In particular, the .01 original-economy target is unchanged and is not attained. A certified fixed-witness floor and actor–critic interchange show why neither further arithmetic precision nor a monotone incumbent alone establishes accurate policy learning.

A second, nonlinear application supplies an explicit accuracy–work result on a genuinely vector-valued state. A heterogeneous multi-product inventory planner has 12 decision dates, coupled nonquadratic costs, and up to 128 state coordinates. The neural map takes the full initial state and returns a control plan, without optimal-value or optimal-control labels. Strong convexity yields a state-uniform certificate for exact-real gradient corrections to that plan. At 128 coordinates, the total-cost bound falls from 0.124820 after 16 corrections to 0.000359609 after 24 and below 1.037×10^{-6} after 32. This is an unknown-value nonlinear planning problem, not the two-mode Riccati example retained for validation. Zero-start gradient, accelerated-gradient,

and quasi-Newton alternatives are evaluated with the same independent cost-loss certificate. They remain strong competitors; the experiment establishes an accurate neural-initialized solver, not a demonstrated neural speed advantage.

Economic interpretation also requires keeping numerical objects distinct. The sharp deterministic time-control/dual library in the original economy is not a neural result. Its price-uniform regret below 0.009982 can be offset by a rigorously checked, budget-financed initial-wealth increment of .0078 at initial wealth 1.25, using an explicitly adjusted feasible consumption policy. This sufficient compensation measure respects the original consumption cap and stopping contract. The earlier externally financed top-up remains a secondary normalization. Neither measure converts the much larger original neural certificates into sharp welfare claims.

25.1. *Contribution and relation to the literature*

Stopped verification, Bellman contraction, strong-convexity error bounds, and gradient contraction are classical ingredients. Judd (1998) and Kushner and Dupuis (2001) develop numerical dynamic programming and controlled Markov-chain approximation. Barles and Souganidis (1991) establishes a fundamental monotone-approximation framework, and Brumm and Scheidegger (2017) develops adaptive sparse-grid methods for computational economics. Information and control relaxations provide dual comparisons (Brown, Smith, and Sun, 2010, Brown and Smith, 2011). Neural dynamic-control and differential-equation methods include Han and E (2016), Han, Jentzen, and E (2018); the historical external comparison uses Cai, Fang, and Zhou (2025).

Our contribution is the combination of an identified stopping-trace inconsistency, an executed complete-cover neural objective with a checkable policy-loss implication, and a nonlinear control specialization with an explicit state-uniform work-accuracy bound. The new objective theorem does not claim global convergence of Adam, and the nonlinear specialization does not claim to remove the difficulty of arbitrary stochastic high-dimensional Bellman equations. The original-economy upper comparison always retains the full control opportunity. The arithmetic implementation uses MPFR (Fousse et al., 2007); shared analytic formulas, stopping arguments, and program logic remain part of its trusted mathematical base.

The paper presents the model and verification foundations first, followed by the stopping correction, the certificate-aligned algorithm, the original-economy evidence, and the nonlinear control application. Historical coefficient transport, state-price continuation, external holdouts, adverse training outcomes, and specialized dual calculations remain available in the main text and supplement. All new numerical statements are tied to immutable source and result objects; all 60 prescribed original-economy checkpoints are included in the new objective audit.

26. CONCLUSION

The objective of neural Bellman operators is an accurate state-contingent solution of the original economic decision problem. The method developed here makes the connection between training and that objective explicit. A complete-cover signed residual envelope guides neural proposals, independent directed arithmetic determines acceptance, and a quantitative envelope theorem states what the accepted object certifies. Stopping accessibility corrects a genuine inconsistency of all-face hard-settlement critics without changing the economy.

The revised evidence separates numerical progress from statements that the data cannot support. The complete-cover objective produces decreasing raw certificates at all prescribed budgets in ten original-economy trajectories, reversing the earlier sampled-objective anti-alignment. Matched boundary ablations, unrestricted classical candidates, method-specific witnesses, and

a computable fixed-witness floor identify the remaining obstruction. The original .01 neural target is still not met, and monotone certificate descent is not presented as monotone true policy improvement.

The nonlinear multi-product application supplies a different, positive accuracy–work result: a full-state neural initialization with a provably contracting correction attains a state-uniform total-cost bound below 1.037×10^{-6} at 128 coordinates. Strong classical alternatives remain competitive under the same certificates. A separately identified original-economy primal/dual library admits explicit budget-financed wealth compensation. These results preserve the neural-solver objective while making every theorem, numerical claim, comparison, and welfare interpretation refer to the exact object for which it is established.

27. HISTORICAL R19 EXPOSITION, PRESERVED VERBATIM

The following two sections retain the preceding manuscript’s exposition. They are historical and do not override the R21 result domains.

28. INTRODUCTION

A numerical policy and an error bound for that policy are different economic objects. An approximation to a value function can improve while the associated consumption, investment, or adjustment decision remains unchanged. This distinction matters for neural dynamic programming: a training criterion that reduces an upper bound on regret need not increase the payoff of the delivered policy. A useful numerical method must identify both changes rather than treating either one as evidence for the other.

We develop Neural Bellman Operators with value-separated policy acceptance. A neural network proposes controls. A policy evaluator supplies a lower and an upper bound for the payoff of those controls under the original stopping contract. An optimal-comparison witness supplies a separate upper bound on the value of the unrestricted economic problem. A policy update is accepted only when the new payoff lower bound exceeds the incumbent payoff upper bound on the declared comparison set. Updating the optimal-comparison witness does not, by itself, change the incumbent policy. The resulting distinction is operational: the acceptance test concerns an actual policy-value difference, whereas the optimality test concerns the distance from that policy to the original optimum.

Our main new analytical result makes policy-sensitive comparison useful beyond a single initial condition. For budget-feasible time controls in the original stochastic economy, a coupled-payoff calculation bounds the variation of a policy-value difference across an initial-state rectangle. It uses differences between the two policies, rather than adding separate, much larger value-function Lipschitz constants. Original CRRA utility, common Gaussian shocks, exact terminal-wealth accounting, and explicit stopping corrections enter the calculation. An independently enclosed value difference at the reference state therefore yields a uniform policy-improvement certificate on a continuum of initial states. A related, observable progress inequality certifies contraction of true policy regret without substituting a critic loss for that regret.

The numerical implementation trains new neural control generators from model-only random starts. No inherited optimal policy, dual witness, or solution label enters their training objective. Five seeds, four prescribed work budgets, and the independent payoff acceptance rule are fixed before execution. All fifteen retained transitions increase certified policy values at the reference state. Initial-to-final improvement is uniform on $K = [1.98, 2.02] \times [1.24, 1.26]$: the gain is at least 0.016513 utility units for every reported seed. Combining this gain with the independently certified final optimality bound proves that true policy regret decreases by at least 58 percent throughout K . At the reference state it decreases by at least 75 percent. These are statements about the economic payoff, not about changes in a residual envelope.

The final reference-state optimality bounds lie between 0.009756577 and 0.009774061, below the unchanged .01 target. The corresponding initial-state-uniform bound is 0.011816625. Each result concerns newly generated neural time controls, held until the original first exit, with a budget-preserving dependence on initial wealth. The optimal upper comparison retains all originally admissible adapted controls. Neither this policy class nor its initial-state rectangle is identified with the full-state feedback architecture. The latter’s best retained complete-domain certificate remains approximately 7.27832. The original objective of an accurate solution over the entire state–time domain is preserved; a localized certificate is not presented as its completion.

Continuation values are the second substantive issue. The full-state feedback experiments learn critics whose upper-wealth traces remain close to immediate settlement, even though

continuation is valuable when that face is inaccessible to the candidate. We evaluate the new policies' one-sided continuation values independently. Their certified excess over immediate settlement exceeds 6.708771 uniformly on the tested preference segment. These evaluations provide policy-induced targets for a continuation trace without using the unknown optimal value as a label. A separate, complete-cover intervention raises the old critic's trace while holding its actor fixed. Its global residual bound worsens. The distinction is informative: removing a necessary trace obstruction is not sufficient to construct a globally accurate Bellman upper witness.

The paper also quantifies the contribution of learned initialization in the nonlinear multi-product control problem. On a new, predeclared finite query distribution, neural, zero-start, and accelerated starts are compared at the same directed loss tolerances, rather than after an arbitrary common number of corrections. The learned map reduces gradient work in several regimes, particularly at 8 and 32 state coordinates. At 128 coordinates its advantage is confined to a coarser target; acceleration wins at the strictest target. A learned-gradient-dependent theorem explains the relevant correction budget. The existing cube-uniform guarantee remains explicitly initialization-agnostic, and the new finite-query results are not represented as a cover of that cube.

Finally, economic interpretation is attached to the newly certified neural policies themselves. A wealth increase of .0092, financed through an explicit feasible change in consumption, compensates their uniform bound on K . It is 0.736 percent of reference wealth. This result is distinct from the retained compensation result for the non-neural price library and does not normalize the much larger full-domain feedback certificate.

28.1. *Contribution and relation to the literature*

Stopped verification, Bellman contraction, strong-convexity bounds, and interval ordering are classical ingredients. Judd (1998) and Kushner and Dupuis (2001) develop numerical dynamic programming and controlled Markov-chain approximation. Barles and Souganidis (1991) establishes a monotone-approximation framework, and Brumm and Scheidegger (2017) develops adaptive sparse-grid methods for computational economics. Information and control relaxations provide dual comparisons (Brown, Smith, and Sun, 2010, Brown and Smith, 2011). Neural dynamic-control and differential-equation methods include Han and E (2016), Han, Jentzen, and E (2018); the retained external comparison uses Cai, Fang, and Zhou (2025). Directed arithmetic uses MPFR (Fousse et al., 2007).

Our contribution is the connection of an implemented neural policy-value gate to a coupled-payoff continuum certificate, constructive continuation-trace evaluation, and matched-tolerance accounting for learned representations. The five new actors are not inherited deterministic solutions relabeled as neural networks. Conversely, the faster classical transcription is not hidden, and neither interval ordering nor strong convexity alone is claimed as a new general convergence theorem. The complete-cover objective, accessibility theorem, state-price continuation, dual construction, external holdouts, and adverse historical results are retained. Their roles are separated from the new policy-sensitive results throughout.

29. CONCLUSION

Neural Bellman Operators require a separation between learning a policy, evaluating that policy, and bounding the unrestricted optimum. The value-separated update studied here supplies that separation in executable form. New neural actors improve independently certified original-economy payoffs, and a coupled-payoff theorem promotes the improvement to a continuum

of initial states. Their reference-state certificates meet the unchanged .01 target; their state-rectangle certificates admit explicit feasible wealth compensation. These achievements concern the newly generated policies themselves, rather than an inherited non-neural library.

The same separation clarifies continuation learning and computational work. Feasible-policy evaluation supplies verified one-sided continuation targets, but a scalar critic lift is not a global Bellman solution. Learned initialization reduces matched-tolerance gradient work in identified nonlinear-control regimes, but the cube-uniform correction theorem remains available without training and classical acceleration remains a strong comparator. Complete feedback certificates, every historical failure, and all matched classical results are retained. The objective remains an accurate, competitive solution of the original economic problem over its full state–time domain. The new policy-sensitive results advance that objective without substituting a change of witness, a restricted initial set, or a different numerical method for the result being claimed.

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